Build the Recommender

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected implementation record, with a validation extension.

Trace the selector, handle no match, repair ties and explain practice before the independent catalogue.

Keep one habit: Retain code version, accepted/rejected rows, ranked output, expected results and limitations.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your **explanation**.

- Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Project Problem and Success Test

Agents and Tools

Predictive vs Generative

W1 · Trace the main activity tie

> Explain the best-so-far change using the written policy.

Your evidence record

- Main activity example: B(fun3,learning2,minutes10), A(2,4,9), limit10; score2fun+learning; equal scores8 use ascending name. Both orders selectA;limit8or emptyinput returnsNone. Full pipeline uses exact integer ranges, rejects every duplicate-name occurrence and reports invalid rows separately from ineligible rows. Separate animal-book picture uses illustrated/comic points and ID ties; independent mystery task has its own shorter-page tie.

- Why does A win the worked example even though B is **process**ed first?

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Handle no match

> Trace an empty list and the main example under limit8.

Your evidence record

- Main activity example: B(fun3,learning2,minutes10), A(2,4,9), limit10; score2fun+learning; equal scores8 use ascending name. Both orders selectA;limit8or emptyinput returnsNone. Full pipeline uses exact integer ranges, rejects every duplicate-name occurrence and reports invalid rows separately from ineligible rows. Separate animal-book picture uses illustrated/comic points and ID ties; independent mystery task has its own shorter-page tie.

- State recommend([], 10) and the worked items with limit 8.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair order dependence

> Declare the correct tie policy and rerun both orders.

Your evidence record

- Main activity example: B(fun3,learning2,minutes10), A(2,4,9), limit10; score2fun+learning; equal scores8 use ascending name. Both orders selectA;limit8or emptyinput returnsNone. Full pipeline uses exact integer ranges, rejects every duplicate-name occurrence and reports invalid rows separately from ineligible rows. Separate animal-book picture uses illustrated/comic points and ID ties; independent mystery task has its own shorter-page tie.

- A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Explain the additional activity practice

> Keep the supplied X/Y/Z practice distinct from the selected book assessment.

Your evidence record

- Main activity example: B(fun3,learning2,minutes10), A(2,4,9), limit10; score2fun+learning; equal scores8 use ascending name. Both orders selectA;limit8or emptyinput returnsNone. Full pipeline uses exact integer ranges, rejects every duplicate-name occurrence and reports invalid rows separately from ineligible rows. Separate animal-book picture uses illustrated/comic points and ID ties; independent mystery task has its own shorter-page tie.

- Write a one-sentence **explanation** of X winning without calling it objectively best.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Rank the new mystery catalogue independently

> Keep the first attempt before hints.

Your evidence record

- This mystery-book assessment has its own score and tie policy: use the supplied mystery/page requirements,3genre+2illustration points and shorter-page tie. Do not reuse the worked activity name tie or animal-book ID tie. The supplied rows are fictional and complete for this comparison; no live lookup or new preference estimate is required.

- Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated.

  Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	

Evidence	My record
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Handle no match for Implementing a **recommender**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Implementing a **recommender**. Proposed **explanation**: "A successful run proves the conclusion". Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and **evidence** to display.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 33 Test and reflect

▼ Extra practice · A revealing test

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

My explanation after practice

Record support used: none, hint or teacher **explanation**.

▼ **Reference · Selected W4 response copy**

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

My reasoning and evidence

▼ Extension · Building the data pipeline — W1 Trace a validation

REFERENCE · SECTION 17

Extension · Building the data pipeline — W1 Trace a validation

> Retained extension task with its own teacher key and response.

Extension question and record

- Explain why a row with fun "4" fails while fun 4 can pass.

My reasoning and evidence

▼ Extension · Building the data pipeline — W2 Check missing fields

REFERENCE · SECTION 18

Extension · Building the data pipeline — W2 Check missing fields

> Retained extension task with its own teacher key and response.

Extension question and record

- What does valid_item return if learning is absent?

My reasoning and evidence

▼ **Extension · Building the data pipeline — W3 Repair the connection**

REFERENCE · SECTION 19

Extension · Building the data pipeline — W3 Repair the connection

> Retained extension task with its own teacher key and response.

Extension question and record

- A program validates rows but still sends the original raw list to recommend. What should change?

My reasoning and evidence

▼ **Extension · Building the data pipeline — W4 Build independently**

REFERENCE · SECTION 20

Extension · Building the data pipeline — W4 Build independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Combine validation and the selector in this week's lesson for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and **explanation**.

My reasoning and evidence

▼ **Extension · Building the data pipeline — W5 Handle duplicates**

REFERENCE · SECTION 21

Extension · Building the data pipeline — W5 Handle duplicates

> Retained extension task with its own teacher key and response.

Extension question and record

- Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional book chooser filters a request before comparing preference scores.

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

AI ACADEMY / AI-2/33

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/33

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Filter

A,B,D meet subject and page requirements; C fails subject.

2. Compute scores

A=2; B=1; D=3.

3. Choose best

Return D with both preference matches recorded.

4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B does not win after D is excluded even though it has fewer pages.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Transparent Book **Recommender**

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional book chooser filters a request before comparing preference scores.
- Which labelled step matches this detail? Reordering the **input** rows must not change D's unique winning score.
- 1. Choose best
- 2. Filter
- 3. **Deterministic output**
- 4. Compute scores
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the **stage** that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after trying

After your attempt, compare with the supported **explanation**: **Deterministic output**
Reordering the **input** rows must not change D's unique winning score. Eligibility and ranking answer different questions.

C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner.

Save rejected reasons for an **explanation**, not just the final ID.

Changed-case question

- Changed-case practice: Change D's pages to 91, leaving every other field unchanged. Which conclusion is supported by the supplied facts?
- 1. D becomes ineligible, so A wins with score 2.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported **explanation**: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Repair and keep evidence

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

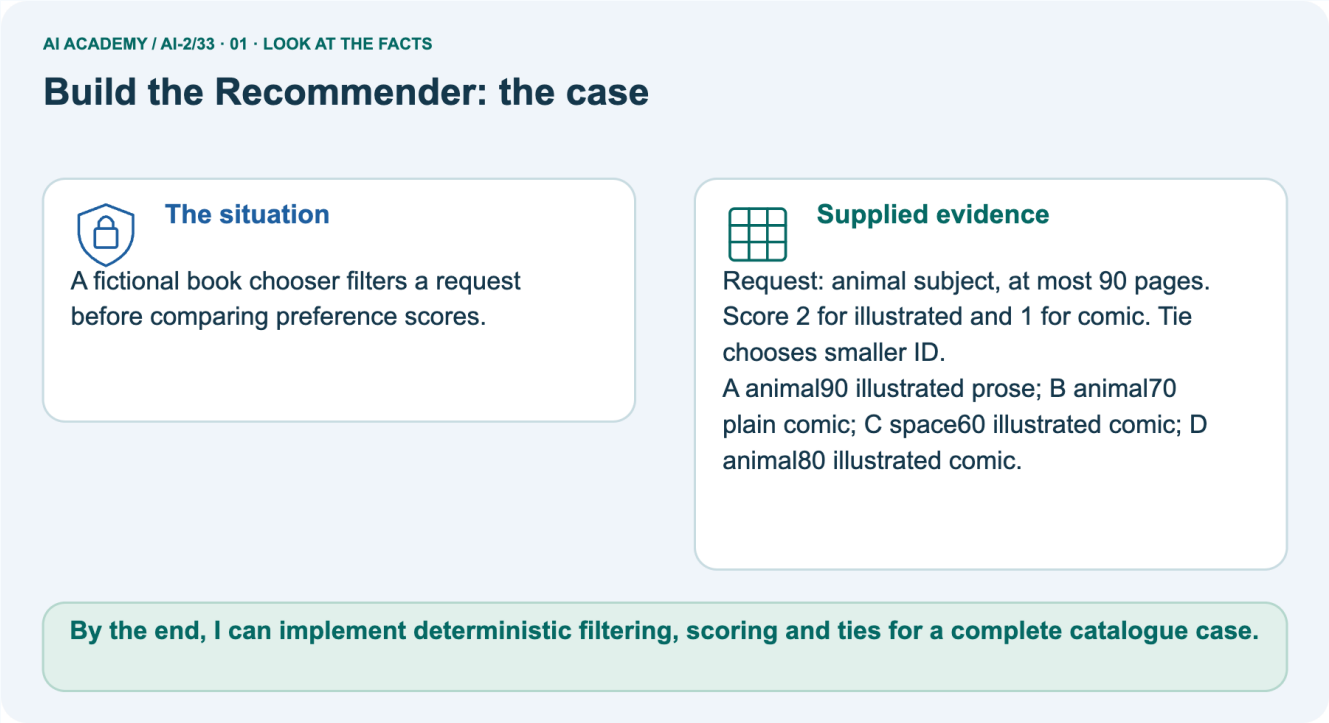
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Build the Recommender: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

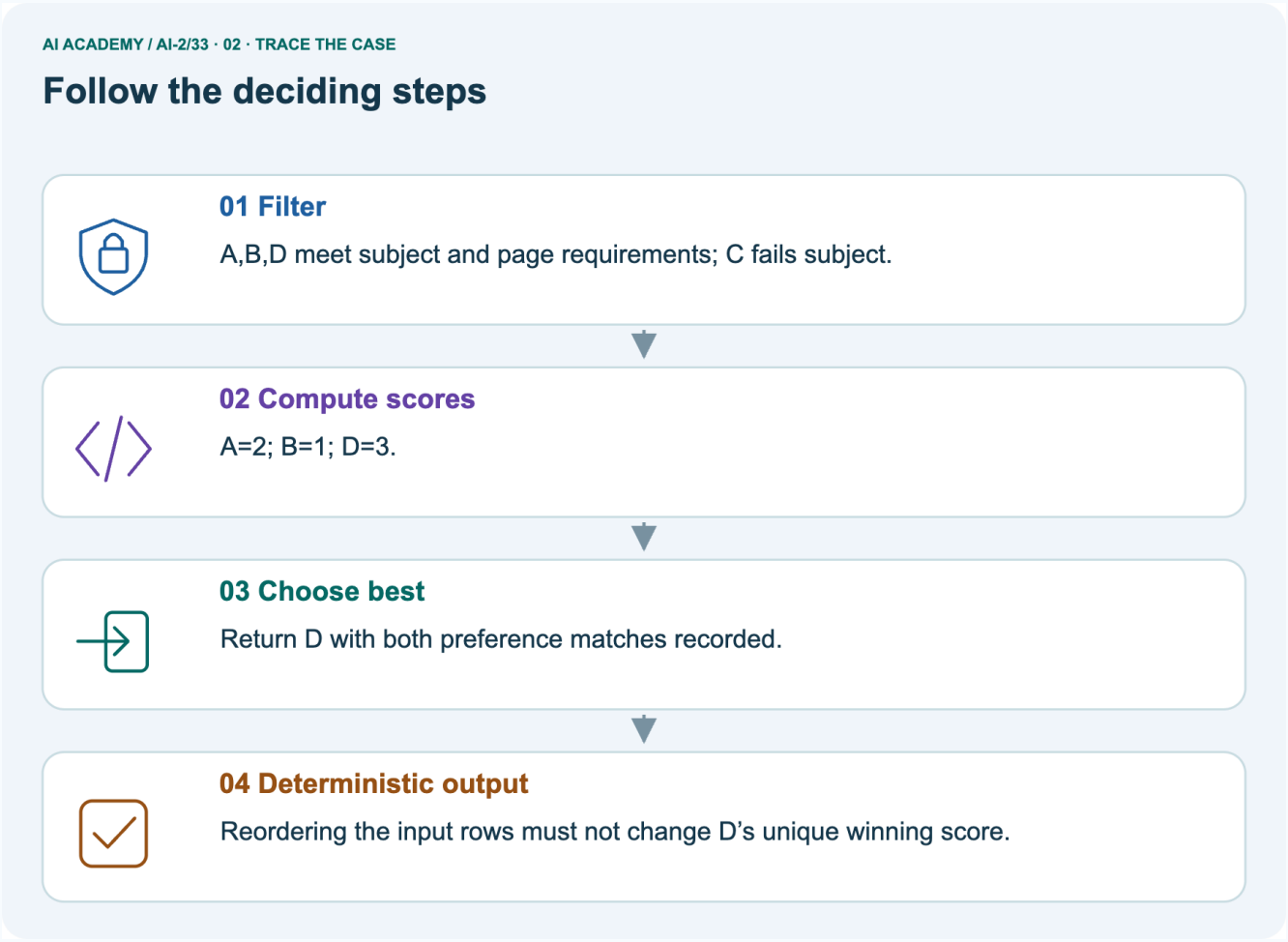
A fictional book chooser filters a request before comparing preference scores. Supplied facts: Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Filter: A,B,D meet subject and page requirements; C fails subject. → Compute scores: A=2; B=1; D=3. → Choose best: Return D with both preference matches recorded. → **Deterministic output**: Reordering the **input** rows must not change D's unique winning score.

Point to each numbered **stage** and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Filtering before scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

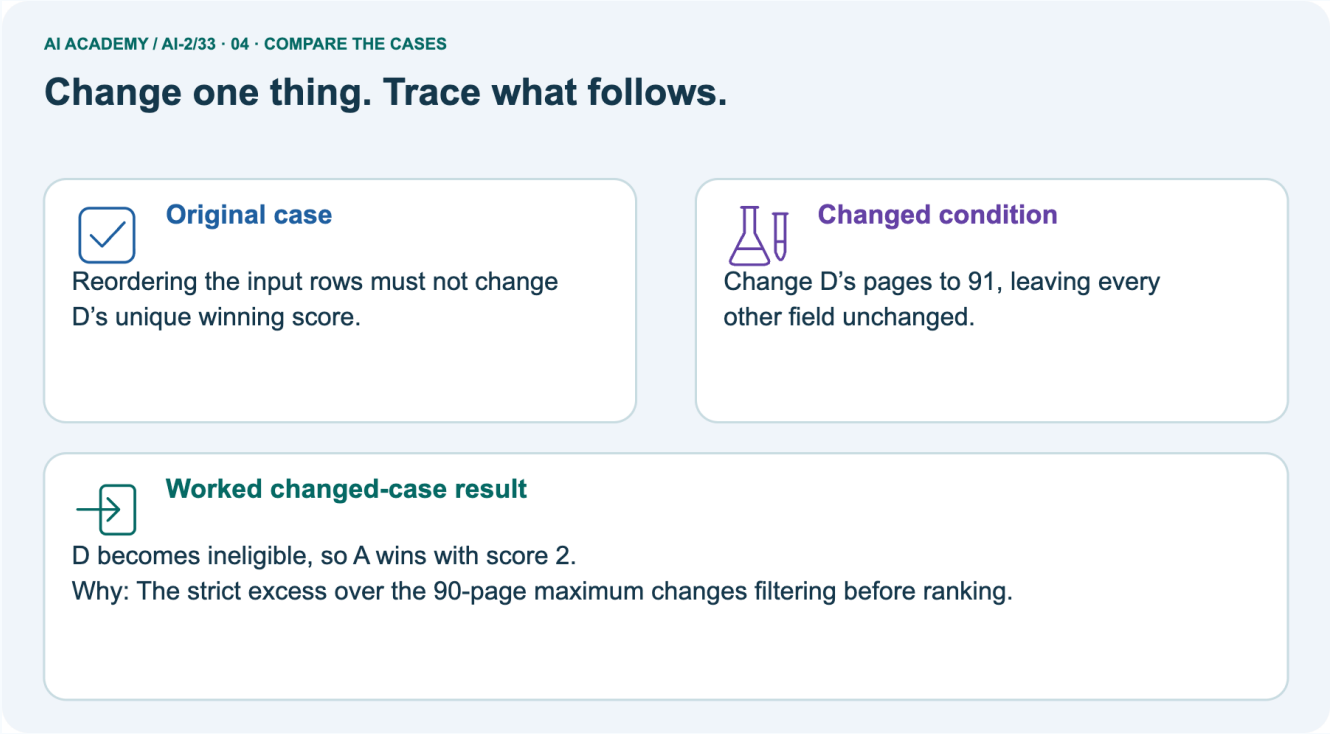
Filtering before scoring. A: Pages = 90, Preference score = 2, Status = eligible. B: Pages = 70, Preference score = 1, Status = eligible. C: Pages = 60, Preference score = 3, Status = subject fails.

D: Pages = 80, Preference score = 3, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

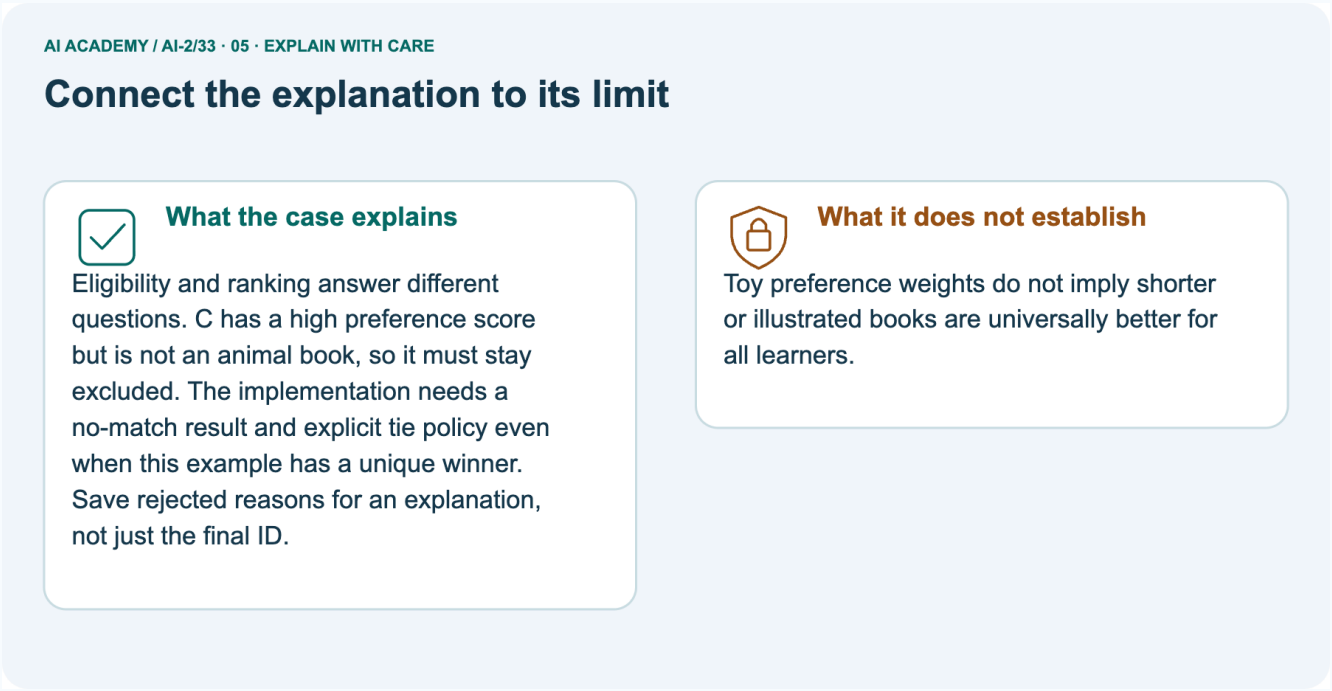
Original case: Reordering the **input** rows must not change D's unique winning score. Changed condition: Change D's pages to 91, leaving every other field unchanged. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

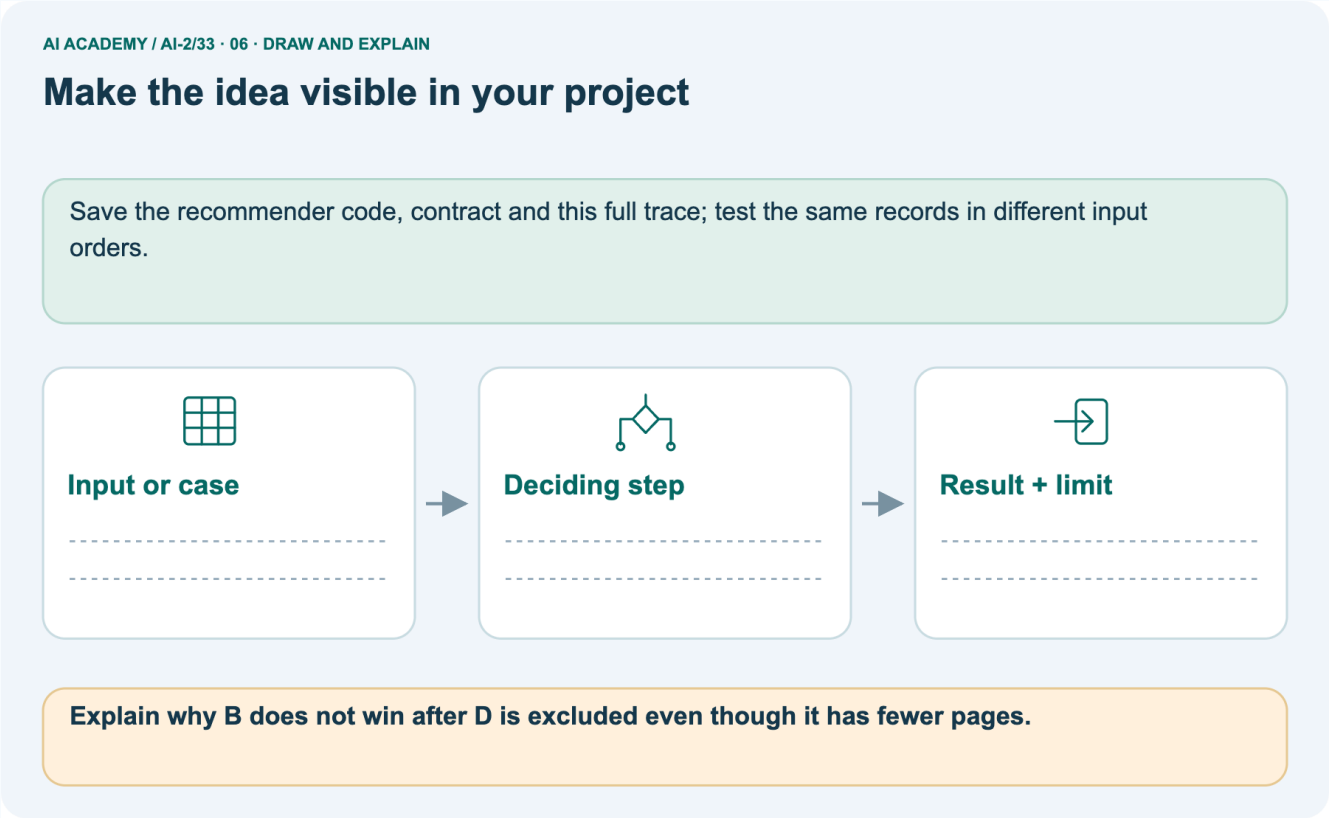
The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

Limit: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B does not win after D is excluded even though it has fewer pages.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Build the Recommender: the case

02. Follow the deciding steps

03. Filtering before scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.